

Analysis of Variance

CSU, Chico Math 314

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outline

Recap

ANOVA

- definition

- intuition

- assumptions

- test statistic

- F-distribution

Example

ANOVA Details

- fitted values

- residuals

Take Away

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recap, Two Sample t -test

The two sample t -test compares the means of two groups in the hypothesis:

$$H_0 : \mu_1 = \mu_2$$

$$H_A : \mu_1 \neq \mu_2.$$

This test required a new statistic with a new calculation for standard error.

recap, Two Sample t -test

The two sample t -test formulas

$$t_{df} = \frac{(\bar{X}_a - \bar{X}_b) - (\mu_a - \mu_b)}{s_{\bar{X}_a - \bar{X}_b}} \quad \text{and} \quad (\bar{X}_a - \bar{X}_b) \pm t_{df}^* s_{\bar{X}_a - \bar{X}_b}$$

where

$$s_{\bar{X}_a - \bar{X}_b} = \sqrt{\frac{s_a^2}{n_a} + \frac{s_b^2}{n_b}}$$

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Analysis of Variance, motivation

If there were three or more groups, the two sample t -test would not work. We could force the test on the data by comparing two groups at a time, but this has dangerous implications¹. We thus require a new statistical method, **analysis of variance**, or **ANOVA**.

¹We'll go into more detail on this in the next lecture: Decision Making is Hard

Analysis of Variance, definition

ANOVA

Analysis of variance uses a single hypothesis to check whether the *means* across two or more groups are equal, and uses a new test statistic called F to evaluate this hypothesis.

ANOVA, hypotheses

The ANOVA hypotheses for k groups are

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$$

H_A : at least one mean is different.

ANOVA, hypothesis examples

With ANOVA you can compare the means by groups for many different data sets.

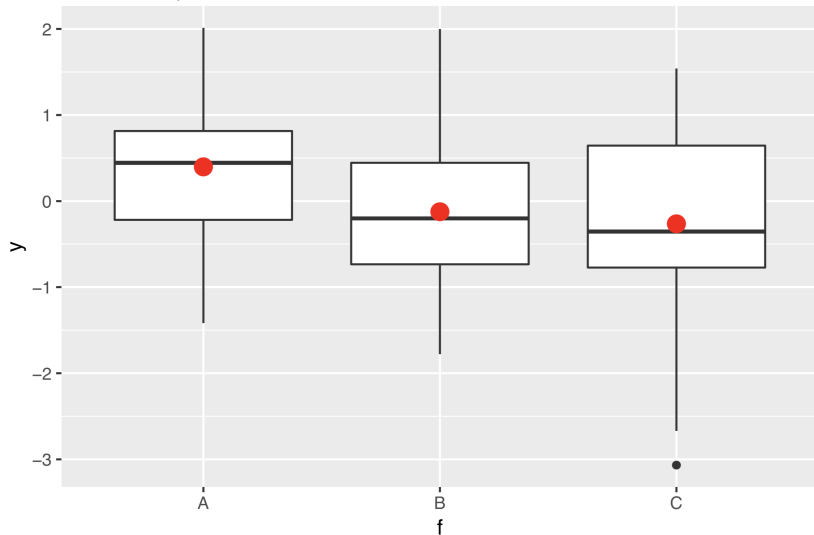
- ▶ mean batting average by position,
- ▶ mean movie budget by genre, (or year),
- ▶ mean CO₂ intake by treatment, or location, (or concentration)
- ▶ mean REM sleep by conservation status,
- ▶ mean birth/body weight by family,
- ▶ ... numerical variable by levels of a categorical variable ...
- ▶ ...

ANOVA, intuition

Analysis of variance tells us about means by groups, despite its name. Large variation **among** the groups relative to small variation **within** the groups indicates different population means.

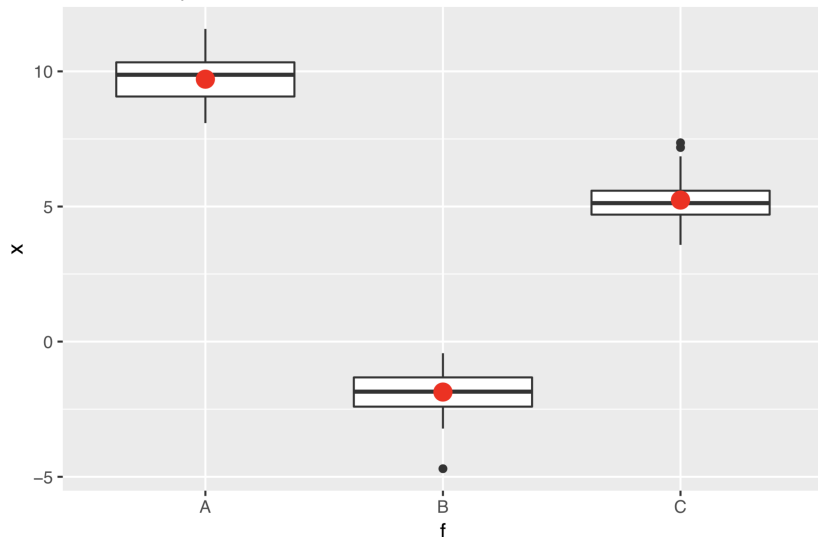
ANOVA, intuition via picture

What do you think of these means (think of variation within and among groups)?



ANOVA, intuition via picture

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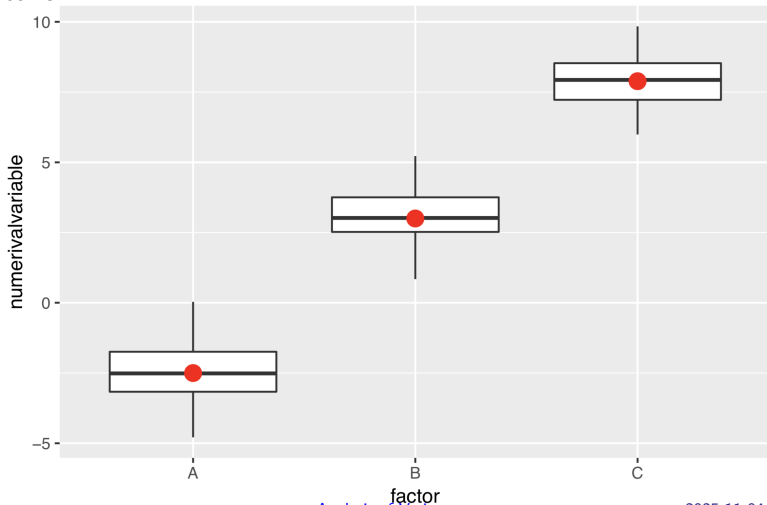
ANOVA, assumptions

The ANOVA has three basic assumptions,

- ▶ independent observations (among and within groups)
- ▶ data within each group are nearly normal, and
- ▶ the variability of each group is about equal.

ANOVA, picture

The last two assumptions can help us visualize just what is going on. It starts with box plots by groups—what other indicators may reveal whether the groups in question have similar or different means?



ANOVA, test statistic

ANOVA calculates one fraction based on two numbers, variation among (between) groups and variation within groups. These two numbers are generally referred to as mean square values.

mean squared variation among groups

MSG is a strictly positive measure of the variation across all groups, and has $df_G = k - 1$ where k represents the number of groups.

mean squared error

MSE is a strictly positive measure of the variation within groups, and has $df_E = n - k$ where k represents the number of groups and n is the sample size.

ANOVA, test statistic

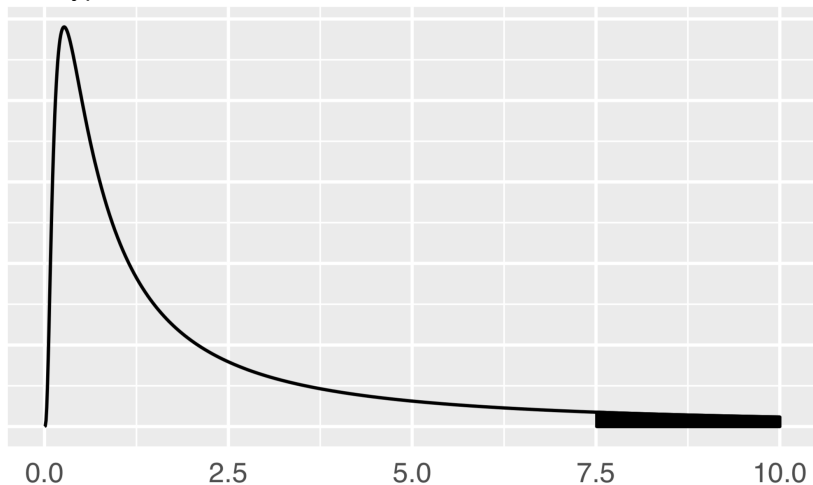
The test statistic of ANOVA follows an F -distribution,

$$F = \frac{MSG}{MSE},$$

and has degrees of freedom associated with the numerator and denominator, df_G and df_E , respectively.

F-distribution

The F -distribution is a probability density function over non-negative numbers. P-values are strictly calculated from the right tail – hence **large** F statistics indicate evidence **against** the null hypothesis.



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ANOVA, example I

Are baseball players paid on average differently by position?

$$H_0 : \mu_{catcher} = \mu_{dh} = \mu_{first} = \dots = \mu_{third}$$

H_A : at least one mean salary is different.

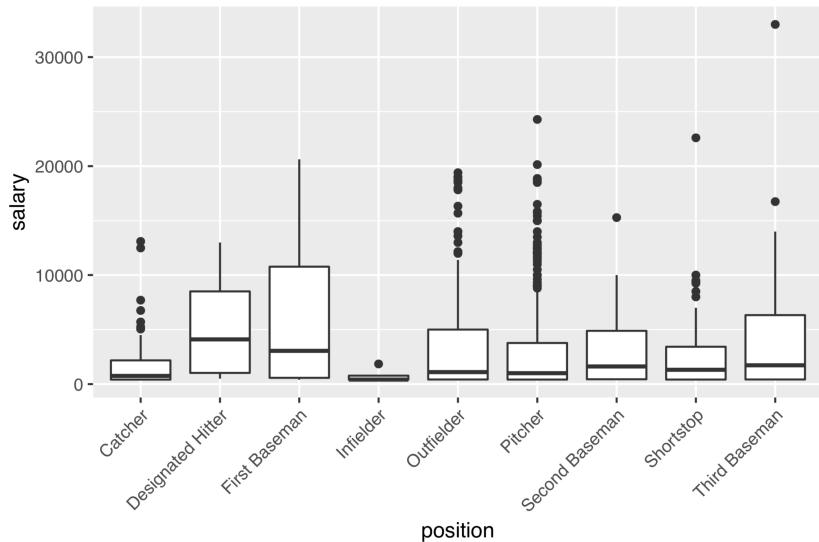
with $\alpha = 0.05$.

ANOVA, example I

Plot the data!

```
suppressMessages(library(ggplot2))
mlb <- read.csv("https://raw.githubusercontent.com/roualdes/data/master/mlb.csv")
p <- ggplot(data=mlb, aes(x=position, y=salary)) +
  geom_boxplot() +
  theme(axis.text.x=element_text(angle=45, hjust=1))
```

ANOVA, example



ANOVA, in R

The R code to run ANOVA is listed here. The `lm` function creates a fitted linear model from the given variables and data set, while the `anova` function performs analysis of variance on the resulting linear model object. You can read more about these functions with `?lm` and `?anova`.

```
model <- lm(salary~position, data=mlb)
anova(model)
```

ANOVA, in R

The tilde $\sim$ indicates to predict the left hand side with the right hand side. Hence, we read the following as "predict salary by different levels of position."

```
model <- lm(salary~position, data=mlb)
```

We thus need to ensure that the variable `position` is a categorical/factor variable.

```
is.factor(mlb[, "position"])  
  
## [1] TRUE
```

ANOVA, example output

The F -statistic and p-value are the most important pieces of information to extract from an ANOVA table. Print output tables from R with

```
anova(model)

## Analysis of Variance Table
##
## Response: salary
##           Df      Sum Sq  Mean Sq
## position    8 6.0975e+08  76219146
## Residuals 819 1.5881e+10 19390502
##           F value    Pr(>F)
## position    3.9307 0.0001422 ***
## Residuals
## ---
## Signif. codes:
##  0 '***' 0.001 '**' 0.01 '*' 0.05
##  '.' 0.1 ' ' 1
```

ANOVA, example conclusion

Because the p-value = $10^{-4} < \alpha = 0.05$ we reject the null hypothesis in favor of the alternative hypothesis. There is sufficient evidence to claim that the mean salary of baseball players varies by position.

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ANOVA

Let's consider the variables Family and average brain weight SB from the dataset `ape::carnivora`.

```
suppressMessages({library(ape)
  library(dplyr)
  library(ggplot2)
  data(carnivora)})
## This will load the data
```

ANOVA

We'll filter down to just five families. ("!" is removing the three families we won't be analyzing)

```
data(carnivora)
carnivs <- filter(carnivora,
                  !(Family %in% c("Ailuridae",
                                   "Procyonidae",
                                   "Viverridae")))
```

ANOVA

Then run ANOVA.

```
mod <- lm(SB~Family, data=carnivs)  
anova(mod)
```

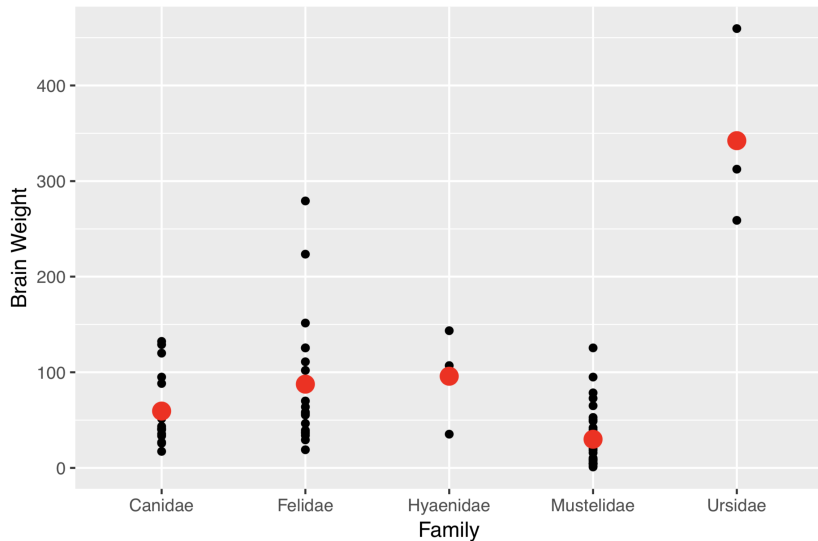
ANOVA, details

ANOVA is all about breaking up the response variable, in general denoted by Y , into component pieces made up of the explanatory variable(s). The slide above breaks up the response variable **brain weight** into an intercept term, and then group means (of sorts) for the remaining $k - 1$ levels of Family.

ANOVA, details graphed

```
p <- ggplot(carnivs, aes(Family, SB)) +  
  geom_point() +  
  stat_summary(fun=mean, colour="red",  
               geom="point", size=4) +  
  labs(y="Brain Weight")
```

ANOVA, details graphed



ANOVA, model

The general ANOVA model is²

$$Y_i = \alpha + \beta_1 X_{i,1} + \dots + \beta_{k-1} X_{i,k-1} + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2),$$

for $i = 1, \dots, n$. Details

- ▶ α is called the intercept
- ▶ $X_{i,j}$ equals 1 or 0 dependent on the i th observation
- ▶ $k - 1$ levels of the categorical explanatory variable X_j get their own estimated value β_j
 - ▶ β_j is difference between the respective group mean and the intercept

²This is the most common model used to fit ANOVA, but it is hardly the most intuitive.

ANOVA, details in R

Calling the function `coefficients` on the linear model returns the intercept and $k - 1$ values β_j . Each group's mean is actually the sum, $\hat{\alpha} + \hat{\beta}_j$ relative to the specific group, j , of interest.

```
coefficients(mod)

##      (Intercept)      FamilyFelidae
##      59.45556      28.14444
## FamilyHyaenidae FamilyMustelidae
##      36.44444      -29.47056
##      FamilyUrsidae
##      282.86944
```

ANOVA, details in R

Compare coefficients to the group means

```
carnivs %>%  
  group_by(Family) %>%  
  summarise(m=mean(SB))  
  
## # A tibble: 5 x 2  
##   Family      m  
##   <fct>    <dbl>  
## 1 Canidae    59.5  
## 2 Felidae    87.6  
## 3 Hyaenidae  95.9  
## 4 Mustelidae 30.0  
## 5 Ursidae   342.
```

ANOVA, details in math

The model predicts the mean value of Y_i , $\mu_{Y|X}$, dependent on the group that observation Y_i came from. For instance, suppose observation i is a member of Felidae,

$$\begin{aligned}\hat{Y}_i &= E(Y_i|X) = E(Y_i|\textit{Felidae}) \\ &= \hat{\alpha} + \hat{\beta}_1(X_{i,1} == 1) + \dots + \hat{\beta}_{k-1}(X_{i,k-1} == 0) \\ &= \hat{\alpha} + \hat{\beta}_1.\end{aligned}$$

ANOVA, fitted values

The **predicted/fitted** values of the model are the group means; the values the model would predict for each group

$$\hat{y}_{ij} = \hat{\alpha} + \hat{\beta}_j.$$

```
yhat <- fitted(mod) # retrieve fitted values
```

ANOVA, details in math

The model says that each observation is the sum of the intercept, the appropriate β_j , and some left over piece, called the **residuals**, denoted by ϵ_i ,

$$y_i = \alpha + \beta_1 X_{i,1} + \dots + \beta_{k-1} X_{i,k-1} + \epsilon_i.$$

The **residuals** make up the difference between the observation and the sum of the intercept and each β_j .

ANOVA, residuals

residuals/errors

The theoretical (or random) residuals/errors are often denoted by ϵ_i in statistical models. Once observed, after fitting the model, we define the residuals as $e_i = y_i - \hat{y}_i$.

```
residuals(mod) # retrieve residuals
```

Note

The difference in the notation between ϵ_i and e_i is similar to that with random variables: X denotes a random variable, not yet observed, and x is the no longer random value that X took on. Here, e_i denotes the observed value of the random variable ϵ_i .

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Take away

- ▶ ANOVA breaks up the mean of the response variable into components
 - ▶ components are specified by a categorical explanatory variable
- ▶ Normality assumption is only necessary for small sample sizes
 - ▶ interest lies in means $\Rightarrow$ Central Limit Theorem
- ▶ equal variance assumption qualitatively checked with plots
 - ▶ there exist more formal ways to check this ...
- ▶ multiple comparisons is a silent killer